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Information processing apparatus, information processing method, program, and information processing system

Abstract

The information processing apparatus includes a processor and a memory which stores a program to be executed by the processor. The information processing apparatus acquires position and attitude information representing a position of a camera mounted on a flying object and an attitude of the camera, acquires camera configuration information including an optical configuration of the camera, calculates an imaging range of the camera based on the position and attitude information and the camera configuration information, acquires a survey range map showing a survey range, acquires survey target house information including a position of a survey target house in the survey range map, determines whether or not the survey target house is included in the imaging range, specifies a not-yet-imaged house that does not have a history of being determined to be included in the imaging range, and superimposes information on the not-yet-imaged house on the survey range map.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a Continuation of PCT International Application No. PCT/JP2022/011197 filed on Mar. 14, 2022 claiming priority under 35 U.S.C § 119(a) to Japanese Patent Application No. 2021-048926 filed on Mar. 23, 2021. Each of the above applications is hereby expressly incorporated by reference, in its entirety, into the present application.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to an information processing apparatus, an information processing method, a program, and an information processing system.

2. Description of the Related Art

(2) In a flight imaging system in the related art, an image obtained from a camera mounted on a flying object is displayed on a monitor installed on a ground, and, through the image displayed on the monitor, an imaging person performs imaging while checking the image.

(3) JP1999-187363A (JP-H11-187363A) discloses a mobile station video transmission system comprising a mobile station on which a camera is mounted and a reception station that receives a captured image of the camera transmitted from the mobile station. The system disclosed in JP1999-187363A (JP-H11-187363A) generates a map image showing a position and an imaging range of the mobile station, and the like by using the captured image of the camera transmitted from the mobile station and incidental data of the captured image, and displays the generated map image on a monitor.

(4) The imaging range of the mobile station is generated by computing four corner coordinates of the imaging range from position data of a helicopter constituting the mobile station, attitude angle data of the helicopter, orientation direction data of the camera, and a zoom ratio of the camera.

(5) JP2020-1586A discloses a structure deterioration inspection support system that generates a three-dimensional model of a structure from an image group obtained by imaging the structure from a plurality of viewpoints and that displays the three-dimensional model of the structure.

(6) In the system disclosed in JP2020-1586A, on an individual screen in which an image selected by a user from a plurality of images is displayed, the image is displayed such that the user can see a portion of a not-yet-imaged range of the structure. Accordingly, the user can recognize a deterioration inspection omission portion in the structure.

(7) WO2013/051300A discloses a disaster situation grasping system that grasps a situation of a disaster area based on an image captured aerially from a flying object such as a helicopter. In the system disclosed in WO2013/051300A, in a case in which a disaster occurs, a marker is attached to a damaged structure, and relevant information related to the structure is displayed.

SUMMARY OF THE INVENTION

(8) However, in a housing damage certification survey conducted by a local government, at least one image in which a survey target house is captured is required. The survey target house that is left out of the imaging performed using the camera provided in the flying object is captured by an on-site investigator. In that case, the leave-out of the imaging using the flying object may cause a delay in issuance of a disaster certificate.

(9) The flight imaging system in the related art has difficulty in determining whether or not one or more captured images have been obtained for all the survey target houses. In addition, in a case in which imaging is performed using the camera provided in the flying object, it is difficult to determine a not-yet-imaged house from the image displayed on the monitor. In addition, it is difficult to guide the flying object to the not-yet-imaged house.

(10) In the system disclosed in JP1999-187363A (JP-H11-187363A), the imaging range captured

by the camera is displayed on a map, and a location of a disaster area is specified from the map, but it is difficult to specify the not-yet-imaged house in the disaster area from the map.

(11) The system disclosed in JP2020-1586A can grasp the deterioration inspection omission portion of the structure, but JP2020-1586A does not disclose specification of a not-yet-imaged structure.

(12) The system disclosed in WO2013/051300A can grasp characteristics of a disaster-stricken structure, but it is difficult to specify a not-yet-imaged structure in an imaging area.

(13) The present invention has been made in view of such circumstances, and an object of the present invention is to provide an information processing apparatus, an information processing method, a program, and an information processing system which can support specification of a not-yet-imaged house in a case of imaging a survey range.

(14) An information processing apparatus according to the present disclosure is an information processing apparatus comprising: a processor; and a memory in which a program to be executed by the processor is stored, in which the processor executes a command of the program to acquire position and attitude information representing a position of a camera mounted on a flying object and an attitude of the camera, acquire camera configuration information including an optical configuration of the camera, calculate an imaging range of the camera based on the position and attitude information and the camera configuration information, acquire a survey range map showing a survey range, acquire survey target house information including a position of a survey target house in the survey range map, determine whether or not the survey target house is included in the imaging range, specify a not-yet-imaged house that does not have a history of being determined to be included in the imaging range, and superimpose information on the not-yet-imaged house on the survey range map.

(15) According to the information processing apparatus according to the present disclosure, a not-yet-imaged-house image corresponding to the not-yet-imaged house that does not have a history of being determined to be included in the imaging range of the camera is superimposed on the survey range map. Accordingly, the not-yet-imaged house can be specified.

(16) In the survey range map, as a geographical condition of a survey range, a map of the survey range and an aerial photograph of the survey range can be applied.

(17) An information processing apparatus may comprise a display device, and the processor may transmit a display signal representing the survey range map on which the not-yet-imaged-house image is superimposed, to the display device. In such an aspect, the display device can display the survey range map on which the not-yet-imaged-house image is superimposed.

(18) In the information processing apparatus according to another aspect, the processor superimposes, on the survey range map, a not-yet-imaged-house image corresponding to the not-yet-imaged house.

(19) According to such an aspect, visibility of the not-yet-imaged house can be improved.

(20) In the information processing apparatus according to still another aspect, the processor superimposes, on the survey range map, information on an imaged house having a history in which the survey target house is determined to be included in the imaging range.

(21) According to such an aspect, the imaged house can be specified.

(22) In the information processing apparatus according to still another aspect, the processor superimposes, on the survey range map, an imaged-house image corresponding to the imaged house.

(23) According to such an aspect, visibility of the imaged house can be improved.

(24) In the information processing apparatus according to still another aspect, the processor superimposes, on the survey range map, camera position information representing a position of the camera in the survey range map.

(25) According to such an aspect, visibility of the position of the camera can be improved.

(26) In the information processing apparatus according to still another aspect, the processor superimposes, on the survey range map, first navigation information representing a direction in

which the not-yet-imaged house is located.

(27) According to such an aspect, a direction in which the not-yet-imaged house is located can be specified.

(28) In such an aspect, as the first navigation information, a first navigation image representing the first navigation information can be applied.

(29) In the information processing apparatus according to still another aspect, the processor superimposes, on the survey range map, the first navigation information including information representing the number of the not-yet-imaged houses in the direction represented by the first navigation information.

(30) According to such an aspect, the number of the not-yet-imaged houses in the direction in which the not-yet-imaged house is located can be specified.

(31) In the information processing apparatus according to still another aspect, the processor superimposes, on the survey range map, second navigation information representing a distance and a direction from the imaging range to the not-yet-imaged house of interest.

(32) According to such an aspect, the direction of the not-yet-imaged house of interest and the distance to the not-yet-imaged house of interest can be specified.

(33) As the second navigation information in such an aspect, a second navigation image representing the second navigation information can be applied. The second navigation image can include text information.

(34) In the information processing apparatus according to still another aspect, the processor superimposes, on the survey range map, third navigation information including a wide area map in which an entirety of the survey range map is reduced.

(35) According to such an aspect, a position of a survey map in the entire survey range can be specified.

(36) As the third navigation information in such an aspect, a third navigation image representing the third navigation information can be applied.

(37) In the information processing apparatus according to still another aspect, the processor acquires the position of the camera, an altitude of the camera, an orientation of an optical axis of the camera, and a depression angle of the optical axis, as the position and attitude information.

(38) According to such an aspect, an angle of view of the camera can be specified.

(39) In the information processing apparatus according to still another aspect, the processor determines whether or not the not-yet-imaged house is included in the imaging range in a case in which the imaging range moves according to movement of the flying object.

(40) According to such an aspect, a captured image of the entire survey range can be acquired by performing imaging a plurality of times.

(41) In the information processing apparatus according to still another aspect, the processor stores a history in which the survey target house is determined to be included in the imaging range.

(42) According to such an aspect, the not-yet-imaged house can be specified based on a determination history of the survey target house.

(43) An information processing method according to the present disclosure is an information processing method executed by a computer, the method comprising: acquiring position and attitude information representing a position of a camera mounted on a flying object and an attitude of the camera; acquiring camera configuration information including an optical configuration of the camera; calculating an imaging range of the camera based on the position and attitude information and the camera configuration information; acquiring a survey range map showing a survey range; acquiring survey target house information including a position of a survey target house in the survey range map; determining whether or not the survey target house is included in the imaging range; specifying a not-yet-imaged house that does not have a history of being determined to be included in the imaging range; and superimposing information on the not-yet-imaged house on the survey range map.

(44) According to the information processing method according to the present disclosure, it is possible to obtain the same operation and effect as those of the information processing apparatus according to the present disclosure. Configuration requirements of an information processing apparatus according to still another aspect can be applied to configuration requirements of an information processing method according to still another aspect.

(45) A program according to the present disclosure is a program causing a computer to realize: a function of acquiring position and attitude information representing a position of a camera mounted on a flying object and an attitude of the camera; a function of acquiring camera configuration information including an optical configuration of the camera; a function of calculating an imaging range of the camera based on the position and attitude information and the camera configuration information; a function of acquiring a survey range map showing a survey range; a function of acquiring survey target house information including a position of a survey target house in the survey range map; a function of determining whether or not the survey target house is included in the imaging range; a function of specifying a not-yet-imaged house that does not have a history of being determined to be included in the imaging range; and a function of superimposing information on the not-yet-imaged house on the survey range map.

(46) According to the program according to the present disclosure, it is possible to obtain the same operation and effect as those of the information processing apparatus according to the present disclosure. Configuration requirements of an information processing apparatus according to still another aspect can be applied to configuration requirements of a program according to still another aspect.

(47) An information processing system according to the present disclosure is an information processing system comprising: a flying object; a camera mounted on the flying object; and an information processing apparatus equipped with a processor and a memory in which a program to be executed by the processor is stored, in which the processor executes a command of the program to acquire position and attitude information representing a position of the camera mounted on the flying object and an attitude of the camera, acquire camera configuration information including an optical configuration of the camera, calculate an imaging range of the camera based on the position and attitude information and the camera configuration information, acquire a survey range map showing a survey range, acquire survey target house information including a position of a survey target house in the survey range map, determine whether or not the survey target house is included in the imaging range, specify a not-yet-imaged house that does not have a history of being determined to be included in the imaging range, and superimpose information on the not-yet-imaged house on the survey range map.

(48) According to the information processing system according to the present disclosure, it is possible to obtain the same operation and effect as those of the information processing apparatus according to the present disclosure. Configuration requirements of an information processing apparatus according to still another aspect can be applied to configuration requirements of an information processing system according to still another aspect.

(49) According to the present invention, the not-yet-imaged-house image corresponding to the not-yet-imaged house that does not have a history of being determined to be included in the imaging range of the camera is superimposed on the survey range map. Accordingly, the not-yet-imaged house can be specified.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram showing a configuration example of an information processing system according to an embodiment.

- (2) FIG. 2 is a functional block diagram showing an electric configuration of an imaging drone shown in FIG. 1.
- (3) FIG. 3 is a functional block diagram of a controller shown in FIG. 1.
- (4) FIG. 4 is a functional block diagram of a terminal device shown in FIG. 1.
- (5) FIG. 5 is a block diagram showing a configuration example of hardware of the terminal device.
- (6) FIG. 6 is a flowchart showing a procedure of an information processing method according to the embodiment.
- (7) FIG. 7 is a schematic diagram of an angle-of-view calculation method.
- (8) FIG. 8 is a schematic diagram of imaging range calculation.
- (9) FIG. 9 is an explanatory diagram of a survey range map.
- (10) FIG. 10 is an explanatory diagram of a survey range map on which a not-yet-imaged-house image is superimposed.
- (11) FIG. 11 is an explanatory diagram of transition of a survey range map.
- (12) FIG. 12 is an explanatory diagram of a survey range map according to a modification example.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (13) Hereinafter, preferred embodiments of the present invention will be described with reference to the accompanying drawings. In the present specification, the same components are denoted by the same reference numerals, and duplicate description thereof will be omitted as appropriate.
- (14) [Configuration Example of Image Processing System According to Embodiment]
- (15) [Overall Configuration]
- (16) FIG. 1 is a schematic diagram showing a configuration example of an information processing system according to an embodiment. Hereinafter, an information processing system that processes various types of information acquired in flight imaging will be described by taking, as an example, a housing damage certification survey conducted in a case of issuing a disaster certificate.
- (17) An information processing system 10 shown in FIG. 1 comprises an imaging drone 12, a camera 14, a controller 16, a terminal device 18, and a server 20. The imaging drone 12, the camera 14, the terminal device 18, and the server 20 may perform data communication with each other via a network 22.
- (18) In FIG. 1, an arrow line representing data communication between the imaging drone 12 and the network 22 and an arrow line representing data communication between the camera 14 and the network 22 are illustrated using a single arrow line.
- (19) The imaging drone 12 and the camera 14 may individually perform data communication with the network 22, and data communication between the imaging drone 12 and the network 22 and data communication between the camera 14 and the network 22 may be shared.
- (20) The imaging drone 12 is an unmanned aerial vehicle that is remotely controlled using the controller 16. The imaging drone 12 may have an auto-pilot function of flying according to a predetermined program. The imaging drone 12 described in the embodiment is an example of a flying object.
- (21) The camera 14 is supported on a lower part of a main body 12A of the imaging drone 12 using a gimbal 13, and is mounted on the imaging drone 12. The lower part of the main body 12A of the imaging drone 12 represents a side opposite to a rotor 12B in a case in which a position where the rotor 12B is attached is an upper part of the main body 12A of the imaging drone 12.
- (22) The gimbal 13 rotatably supports the camera 14. The gimbal 13 may rotatably support the camera 14 in each of two rotation axis directions orthogonal to each other.
- (23) The camera 14 comprises an optical system and an image sensor. The optical system includes one or more lenses such as a focus lens. As the image sensor, a CCD image sensor, a CMOS image sensor, or the like can be applied. Note that CCD is an abbreviation for Charge Coupled Device. CMOS is an abbreviation for Complementary Metal-Oxide Semiconductor.
- (24) The optical system and the image sensor are not shown in FIG. 1. An imaging lens included in the optical system is illustrated with reference numeral 14B in FIG. 7. In addition, the image sensor

is illustrated with reference numeral **14A** in FIG. 7.

(25) The camera **14** forms an image of subject light received from an imaging range F on an imaging plane of an imaging element. The imaging element outputs an electric signal representing the received subject light. The camera **14** uses a signal processing circuit to generate an image signal of a subject from the electric signal representing the subject light, and outputs the image signal of the subject. The camera **14** may transmit an image signal representing a live view image to the controller **16**. A display **16A** of the controller **16** may display the live view image.

(26) The camera **14** images a house in a survey range SA from the sky above the survey range SA. The camera **14** performs at least one time of imaging to generate one or more still images for the entire range of the survey range SA. The camera **14** can image a plurality of houses in one time of imaging. One time of imaging is synonymous with generation of one still image.

(27) For example, the camera **14** changes the imaging range F according to movement of the imaging drone **12**, performs imaging for the entire range of the survey range SA, and generates a plurality of still images corresponding to the entire range of the survey range SA.

(28) The controller **16** transmits a control signal in a wireless format to the imaging drone **12** to control an operation of the imaging drone **12**. That is, in response to an operation of the controller **16** by an operator, a control signal representing a rotation direction and a rotation speed of the rotor **12B** is transmitted from the controller **16** to the imaging drone **12**. The imaging drone **12** performs ascending, descending, turning, forward/backward/left/right movement, hovering, and the like based on the control signal transmitted from the controller **16**.

(29) The controller **16** may perform data communication with the terminal device **18** and the server **20** via the network **22**.

(30) The terminal device **18** functions as an information processing apparatus that supports an operation of the imaging drone **12**. The terminal device **18** acquires a survey range map representing geographical information of the survey range SA. The terminal device **18** can superimpose various types of information, such as information representing an imaging history of the camera **14**, on the survey range map.

(31) The terminal device **18** comprises a display **18A**. The terminal device **18** uses the display **18A** to display the survey range map representing the geographical information of the survey range SA. Details of the survey range map will be described below.

(32) The terminal device **18** may have a function of the controller **16**. That is, the terminal device **18** may have a wireless communication function with the imaging drone **12** and a function of executing a control program of the imaging drone **12**.

(33) A computer is applied to the server **20**. A form of the computer may be a personal computer or a workstation. The server **20** acquires various types of information transmitted from each of the imaging drone **12**, the camera **14**, and the terminal device **18**.

(34) Wireless communication in 2.4 gigahertz band or the like can be applied to the network **22**. Any wired communication can be applied to the network **22**. A combination of wireless communication and wired communication may be applied to the network **22**. Any communication standard can be applied to the network **22**. The network **22** may be electrically connected to a computer or the like provided in a survey base or the like in a communicable manner.

(35) [Configuration Example of Imaging Drone]

(36) FIG. 2 is a functional block diagram showing an electric configuration of the imaging drone shown in FIG. 1. The imaging drone **12** comprises a GPS receiver **30**, an atmospheric pressure sensor **32**, an azimuth sensor **34**, a gyro sensor **36**, and a communication interface **38**. Note that GPS is an abbreviation for Global Positioning System.

(37) The GPS receiver **30** acquires a latitude and a longitude of the imaging drone **12**. The atmospheric pressure sensor **32** detects an atmospheric pressure in the imaging drone **12**. The imaging drone **12** acquires an altitude of the imaging drone **12** based on the atmospheric pressure detected using the atmospheric pressure sensor **32**. The latitude, longitude, and altitude of the

imaging drone **12** constitute position information of the imaging drone **12**.

(38) The acquisition of the information on the altitude of the imaging drone **12** includes an aspect of calculating the altitude of the imaging drone **12** from the atmospheric pressure detected using the atmospheric pressure sensor **32**. That is, the acquisition may include the meaning of information generation such as computation, estimation, and calculation.

(39) The azimuth sensor **34** detects a predetermined azimuth of the imaging drone **12** in a forward direction. The imaging drone **12** acquires a traveling direction of the imaging drone **12** based on the azimuth acquired using the azimuth sensor **34**.

(40) The gyro sensor **36** detects a roll angle representing a rotation angle with respect to a roll axis, a pitch angle representing a rotation angle with respect to a pitch axis, and a yaw angle representing a rotation angle with respect to a yaw axis. The imaging drone **12** acquires attitude information of the imaging drone **12** based on the rotation angles acquired using the gyro sensor **36**.

(41) That is, the imaging drone **12** may comprise a computation processing unit that calculates parameters and the like applied to the control of the imaging drone **12** from various types of information acquired using various sensors. A combination of a processor and a memory can be applied to the computation processing unit. Details of the processor and the memory will be described below.

(42) The communication interface **38** controls the wireless communication of the imaging drone **12** performed via the network **22**. The communication interface **38** may comprise a communication terminal corresponding to wired communication.

(43) The imaging drone **12** comprises a battery and a charging terminal for the battery. The imaging drone **12** may comprise a residual amount meter that notifies of a residual amount of the battery. The residual amount meter may transmit the residual amount information of the battery to the controller **16** via the communication interface **38**. The battery, the charging terminal, and the residual amount meter are not shown.

(44) [Configuration Example of Controller]

(45) FIG. **3** is a functional block diagram of the controller shown in FIG. **1**. The controller **16** comprises a user interface **50**, an imaging setting unit **52**, a flight control unit **54**, a sensor data storage unit **56**, an image data storage unit **58**, an imaging control unit **60**, and a communication interface **62**.

(46) The user interface **50** includes an operation member operated by an operator, such as a joystick and a button, and the display **16A**. The operation member outputs an operation signal corresponding to the operation of the operator. The display **16A** displays various types of information related to the control of the imaging drone **12**.

(47) The imaging setting unit **52** generates imaging setting based on the operation of the operator and the plan of the housing damage certification survey, and transmits the imaging setting to the camera **14**. The camera **14** performs imaging based on the imaging setting.

(48) The flight control unit **54** generates flight setting based on the operation of the operator and the plan of the housing damage certification survey, and transmits the flight setting to the imaging drone **12**. The imaging drone **12** flies based on the flight setting.

(49) The sensor data storage unit **56** stores sensor data transmitted from each of the atmospheric pressure sensor **32**, the azimuth sensor **34**, and the gyro sensor **36** provided in the imaging drone **12**. The flight control unit **54** controls the flight of the imaging drone **12** based on the sensor data stored in the sensor data storage unit **56**.

(50) The image data storage unit **58** stores image data transmitted from the camera **14**. The image data may be associated with an imaging date and time, an imaging condition set in the camera **14**, the sensor data, and the like. The display **16A** may display an image represented by the image data stored in the image data storage unit **58**.

(51) The imaging control unit **60** controls still image capturing of the camera **14** based on the imaging setting and the flight setting. The imaging control unit **60** may control the still image

capturing of the camera **14** in response to the operation of the operator. That is, the imaging control unit **60** adjusts an angle of view of the camera **14** at a predetermined imaging position, and causes the camera **14** to perform imaging. The imaging control unit **60** acquires the image data from the camera **14** and stores the acquired image data in the image data storage unit **58**.

(52) The communication interface **62** performs wireless communication with the communication interface **38** provided in the imaging drone **12**. The communication interface **62** may be electrically connected to the network **22** shown in FIG. **1** in a data communicable manner.

(53) [Configuration Example of Terminal Device]

(54) FIG. **4** is a functional block diagram of the terminal device shown in FIG. **1**. The terminal device **18** comprises a survey range map acquisition unit **80**. The survey range map acquisition unit **80** acquires a map of the survey range or geographical information corresponding to the map from the server **20** shown in FIG. **1**.

(55) As the survey range map, a map of the survey range SA that specifies an address or the like of the survey target house can be applied. As the survey range map, an aerial photograph of the survey range SA may be applied, or a processed map image in which components of the map are replaced with schematized images may be applied.

(56) For example, as the survey range map, a map image to which various types of geographical information are added based on the map of the survey range SA or the aerial photograph of the survey range SA can be applied. The geographical information may include text, photographs, illustrations, figures, and symbols.

(57) The acquisition of the survey range map may include an aspect of acquiring the geographical information of the survey range and generating the survey range map based on the geographical information of the survey range. The acquisition of the survey range map may include an aspect of acquiring the map of the survey range and processing the map of the survey range.

(58) A two-dimensional coordinate system is set for the survey range map. As the two-dimensional coordinate system, an orthogonal coordinate system can be applied. As the origin of the two-dimensional coordinate system, any position in the survey range map can be applied. Any position in the survey range map can be defined using coordinate values.

(59) The terminal device **18** comprises a camera attitude information acquisition unit **82**. The camera attitude information acquisition unit **82** acquires sensor information from the imaging drone **12** at a predetermined imaging position, and acquires camera attitude information representing the attitude of the camera **14** at a predetermined imaging position based on the acquired sensor information.

(60) The terminal device **18** comprises a camera configuration information acquisition unit **83**. The camera configuration information acquisition unit **83** acquires camera configuration information including an optical configuration of the camera **14**, such as a focal length of the imaging lens provided in the camera **14** and a size of the image sensor provided in the camera **14**, from the camera **14**.

(61) The terminal device **18** can acquire and store the camera configuration information as a constant in advance. For example, the terminal device **18** may acquire identification information of the camera **14**, and acquire the camera configuration information that matches the identification information from the server **20**.

(62) The terminal device **18** comprises an imaging range calculation unit **84**. The imaging range calculation unit **84** calculates the imaging range F of the camera **14** on the ground in a case in which the imaging drone **12** is located at a predetermined imaging position, based on the camera attitude information acquired using the camera attitude information acquisition unit **82** and the camera configuration information acquired using the camera configuration information acquisition unit **83**.

(63) The imaging range calculation unit **84** calculates coordinate values of the imaging range F of the camera **14** in the survey range map. For example, the imaging range calculation unit **84**

calculates coordinate values of apexes constituting a trapezoidal shape representing the imaging range F of the camera **14**.

(64) The terminal device **18** comprises a survey target house information acquisition unit **86**. The survey target house information acquisition unit **86** acquires survey target house information including the address, shape, and the like of the survey target house. The survey target house information is applied to the specification of the survey target house in the survey range map acquired by using the survey range map acquisition unit **80**.

(65) The survey target house information acquisition unit **86** calculates coordinate values of the survey target houses in the survey range map. The survey target house information acquisition unit **86** calculates coordinate values of apexes constituting a polygonal shape representing the survey target house.

(66) The terminal device **18** comprises an imaged house determination unit **88**. The imaged house determination unit **88** determines the survey target house included in the imaging range F of the camera **14** as the imaged house. The imaged house determination unit **88** stores imaged house information related to the imaged house. The imaged house determination unit **88** stores, as the imaged house information, coordinate values of apexes constituting a polygonal shape representing the imaged house.

(67) The imaged house determination unit **88** stores a history of determination results for all the survey target houses. A storage device provided in the terminal device **18** is applied to store the history of the determination results.

(68) The terminal device **18** comprises a not-yet-imaged house determination unit **90**. The not-yet-imaged house determination unit **90** determines a house that is not included in the imaging range F of the camera **14** as a not-yet-imaged house. The not-yet-imaged house determination unit **90** stores not-yet-imaged house information related to the not-yet-imaged house. The not-yet-imaged house determination unit **90** stores, as the not-yet-imaged house information, coordinate values of apexes constituting a polygonal shape representing the not-yet-imaged house.

(69) The terminal device **18** comprises a survey range map processing unit **92**. The survey range map processing unit **92** performs processing of the survey range map, such as superimposing the not-yet-imaged house information on the survey range map.

(70) The survey range map processing unit **92** may superimpose imaging range information of the camera **14** at the imaging position where the still image capturing is performed on the survey range map. The survey range map processing unit **92** may superimpose the imaged house information on the survey range map.

(71) The survey range map processing unit **92** comprises a not-yet-imaged-house image generation unit **94**. The not-yet-imaged-house image generation unit **94** generates a not-yet-imaged-house image representing the not-yet-imaged house. In addition, the survey range map processing unit **92** may comprise an imaging-range image generation unit **96**. The imaging-range image generation unit **96** generates an imaging-range image representing the imaging range F of the camera **14**. Further, the survey range map processing unit **92** comprises an imaged-house image generation unit **98**. The imaged-house image generation unit **98** generates an imaged-house image representing the imaged house.

(72) The display control unit **99** transmits a display signal representing the survey range map processed as described above, to the display **18A**. For example, the display control unit **99** can transmit a display signal representing the survey range map on which the imaged-house image is superimposed, to the display **18A**.

(73) FIG. 5 is a block diagram showing a configuration example of hardware of the terminal device. The terminal device **18** comprises a processor **100**, a computer-readable medium **102** that is a non-transitory tangible object, a communication interface **104**, and an input/output interface **106**. The terminal device **18** comprises an input device **110** and a display **18A**.

(74) The processor **100** includes a central processing unit (CPU). The processor **100** may include a

graphics processing unit (GPU). The processor **100** is connected to the computer-readable medium **102**, the communication interface **104**, and the input/output interface **106** via the bus **108**. The input device **110** and the display **18A** are connected to the bus **108** via the input/output interface **106**.

(75) The computer-readable medium **102** includes a memory which is a main memory and a storage which is an auxiliary memory. As the computer-readable medium **102**, a semiconductor memory such as a read only memory (ROM) and a random access memory (RAM), a hard disk apparatus, a solid state drive apparatus, or the like can be applied. Any combination of a plurality of devices can be applied to the computer-readable medium **102**.

(76) The hard disk apparatus can be referred to as an HDD which is an abbreviation for Hard Disk Drive in English. The solid state drive apparatus can be referred to as an SSD which is an abbreviation for Solid State Drive in English.

(77) The terminal device **18** is connected to the network **22** shown in FIG. **1** via the communication interface **104**, and is communicably connected to the imaging drone **12**, the server **20**, and the like via the network **22**.

(78) The computer-readable medium **102** stores various programs corresponding to various functions of the terminal device **18**, such as an image processing program **120**, a storage control program **122**, a communication control program **124**, and a display control program **126**.

(79) The image processing program **120** corresponds to processing functions of the survey range map, such as the survey range map acquisition unit **80** and the survey range map processing unit **92** shown in FIG. **5**. For example, the image processing program **120** performs processing of the survey range map acquired by using the survey range map acquisition unit **80**.

(80) In addition, the image processing program **120** generates an image representing various types of information to be superimposed on the survey range map, such as the not-yet-imaged house information, and performs processing of superimposing various types of information on the survey range map.

(81) The storage control program **122** controls storage of various types of data. The storage control program **122** controls storage of image data representing various images generated by using the image processing program **120**. The storage control program **122** controls storage of various types of information acquired by the terminal device **18**.

(82) The communication control program **124** controls communication performed via the communication interface **104**. The display control program **126** controls display of various types of information on the display **18A**.

(83) Various programs stored in the computer-readable medium **102** include one or more commands. The computer-readable medium **102** stores various types of data, various parameters, and the like.

(84) In the terminal device **18**, the processor **100** executes various programs stored in the computer-readable medium **102** to realize various functions in the terminal device **18**. Note that the term “program” is synonymous with the term “software”.

(85) The terminal device **18** performs data communication with an external device via the communication interface **104**. Various standards such as universal serial bus (USB) can be applied to the communication interface **104**. As a communication form of the communication interface **104**, either wired communication or wireless communication may be applied.

(86) The input device **110** and the display **18A** are connected to the terminal device **18** via the input/output interface **106**. An input device such as a keyboard or a mouse is applied to the input device **110**. A touch panel system can be applied to the input device **110**.

(87) As the display **18A**, a liquid crystal display, an organic EL display, or the like can be applied. Any combination of a plurality of devices can be applied to the display **18A**. The term “EL” of an organic EL display is an abbreviation for Electro-Luminescence.

(88) Here, examples of a hardware structure of the processor **100** include a CPU, a GPU, a programmable logic device (PLD), and an application specific integrated circuit (ASIC). The CPU

is a general-purpose processor that executes a program and acts as various functional units. The GPU is a processor specialized in image processing.

(89) The PLD is a processor capable of changing a configuration of an electric circuit after manufacturing a device. An example of the PLD is a field programmable gate array (FPGA). The ASIC is a processor comprising a dedicated electric circuit specifically designed to execute a specific process.

(90) One processing unit may be configured of one of these various processors or may be configured of two or more processors of the same type or different types. Examples of a combination of various processors include a combination of one or more FPGAs and one or more CPUs, and a combination of one or more FPGAs and one or more GPUs. Another example of a combination of various processors includes a combination of one or more CPUs and one or more GPUs.

(91) A plurality of functional units may be configured by using one processor. As an example of configuring a plurality of functional units by using one processor, there is an aspect in which, as typified by a computer such as a client or a server, one processor is configured by a combination of one or more CPUs and software such as a system on a chip (SoC), and the processor is caused to act as a plurality of functional units.

(92) As another example of configuring a plurality of functional units by using one processor, there is an aspect in which a processor that realizes functions of an entire system including a plurality of functional units by using one IC chip is used. Note that the term “IC” is an abbreviation for Integrated Circuit.

(93) As described above, the various functional units are configured by using one or more of the above described various processors as a hardware structure. Furthermore, the hardware structure of the above described various processors is, more specifically, an electric circuit (circuitry) in which circuit elements such as semiconductor elements are combined.

(94) A hardware configuration of the terminal device **18** shown in FIG. **5** can also be applied to a control device, the controller **16**, and the server **20** of the imaging drone **12** shown in FIG. **1**.

(95) [Procedure of Information Processing Method According to Embodiment]

(96) FIG. **6** is a flowchart showing a procedure of an information processing method according to the embodiment. In a survey range map acquisition step **S10**, the survey range map acquisition unit **80** shown in FIG. **4** acquires the survey range map from the server **20**.

(97) In a survey target house information acquisition step **S12**, the survey target house information acquisition unit **86** acquires the survey target house information from the server **20**. A survey target house information display step of displaying the survey target house information acquired in the survey target house information acquisition step **S12** may be executed.

(98) In a position and attitude information acquisition step **S14**, the camera attitude information acquisition unit **82** acquires the position information and attitude information of the camera **14**. In the position and attitude information acquisition step **S14**, the camera attitude information acquisition unit **82** can acquire sensor information from various sensors provided in the imaging drone **12** and calculate a position and an attitude of the camera **14** based on the acquired sensor information. A camera position and attitude information display step of displaying the position information of the camera **14** and the attitude information of the camera **14** which are acquired in the position and attitude information acquisition step **S14** may be executed.

(99) In a camera configuration information acquisition step **S16**, the camera configuration information acquisition unit **83** acquires the camera configuration information of the camera **14** that is stored in advance. In the camera configuration information acquisition step **S16**, camera configuration information stored in the controller **16** or the server **20** may be acquired. A camera configuration information display step of displaying the camera configuration information acquired in the camera configuration information acquisition step **S16** may be executed.

(100) In an imaging range calculation step **S18**, the imaging range calculation unit **84** calculates the

imaging range F of the camera **14** using the position information of the camera **14**, the attitude information of the camera **14**, and the camera configuration information of the camera **14**. An imaging range information storage step of storing imaging range information generated in the imaging range calculation step **S18** may be executed.

(101) In an imaged house determination step **S20**, the imaged house determination unit **88** determines the survey target house included in the imaging range F of the camera **14** calculated in the imaging range calculation step **S18** as the imaged house. An imaged house information storage step of storing imaged house information generated in the imaged house determination step **S20** may be executed. In addition, a determination history storage step of storing a history of determination results in the imaged house determination step **S20** may be performed for all the survey target houses.

(102) In a not-yet-imaged house determination step **S22**, the not-yet-imaged house determination unit **90** determines a not-yet-imaged house that has not been determined as the imaged house in the imaged house determination step **S20** among the survey target houses. A not-yet-imaged house information storage step of storing not-yet-imaged house information generated in the not-yet-imaged house determination step **S22** may be performed. The imaged house determination step **S20** and the not-yet-imaged house determination step **S22** may be executed in parallel.

(103) In a survey range map processing step **S24**, the survey range map processing unit **92** processes the survey range map. In the survey range map processing step **S24**, the not-yet-imaged-house image generation unit **94** generates the not-yet-imaged-house image representing the not-yet-imaged house, and the survey range map processing unit **92** performs processing of superimposing the not-yet-imaged-house image on the survey range map.

(104) In the survey range map processing step **S24**, the imaging-range image generation unit **96** can generate the imaging-range image representing the imaging range F of the camera **14**. In addition, in the survey range map processing step **S24**, the imaged-house image generation unit **98** can generate the imaged-house image representing the imaged house. The survey range map processing unit **92** can perform processing of superimposing the imaging-range image or the like on the survey range map.

(105) In a survey range map display step **S26**, the display control unit **99** transmits the display signal representing the processed survey range map, to the display **18A**. The display **18A** displays the survey range map on which the not-yet-imaged-house image is superimposed.

(106) In an imaging range update determination step **S28**, the terminal device **18** determines whether or not the imaging range F of the camera **14** has been updated. Examples of the example in which the imaging range F of the camera **14** is updated include movement of the imaging position, and re-imaging.

(107) For example, in a case in which the camera attitude information acquisition unit **82** acquires new position information of the camera **14**, the terminal device **18** determines that the imaging range F of the camera **14** has been updated. In addition, in a case in which a user input signal representing the performance of the re-imaging is acquired, the terminal device **18** can determine that the re-imaging is performed.

(108) In the imaging range update determination step **S28**, in a case in which the terminal device **18** determines that the imaging range F of the camera **14** has been updated, the determination result is Yes. In a case in which the determination result is Yes, the process proceeds to the position and attitude information acquisition step **S14**, and steps from the position and attitude information acquisition step **S14** to the imaging range update determination step **S28** are executed until the determination result of No is made in the imaging range update determination step **S28**. In a case in which the imaging range F is updated, the camera configuration information acquisition step **S16** can be omitted.

(109) On the other hand, in the imaging range update determination step **S28**, in a case in which the terminal device **18** determines that the imaging range F of the camera **14** has not been updated, the

determination result is No. In a case in which the determination result is No, a predetermined termination process is executed, and the procedure of the information processing method is terminated.

(110) [Specific Example of Calculation of Imaging Range]

(111) The calculation of the imaging range F of the camera **14** shown in FIG. **1** includes calculation of the angle of view of the camera **14** and calculation of the imaging range F of the camera **14** based on the angle of view of the camera **14**.

(112) FIG. **7** is a schematic diagram of an angle-of-view calculation method. It is assumed that a size of the image sensor **14A** provided in the camera **14** is D, a focal length of the imaging lens **14B** provided in the camera **14** is f, and an imaging distance is L. An angle of view α in a case in which there is no distortion of the imaging lens **14B** is expressed as $\alpha=(D \times L) / f$. The angle of view α here means a length of one side of the imaging range in a case in which the imaging range F is quadrangular, corresponding to a planar shape of the image sensor **14A**. In a case in which the imaging range F is rectangular, lengths of two sides orthogonal to each other are calculated as the angle of view α .

(113) FIG. **8** is a schematic diagram of imaging range calculation. An imaging range of the camera **14** on a ground G is a cross section obtained in a case in which an imaginary quadrangular pyramid, whose apex is a principal point of the imaging lens **14B** provided in the camera **14** and whose perpendicular line is an optical axis A of the imaging lens **14B**, intersects the ground G.

(114) A flight altitude H of the imaging drone **12** corresponds to an imaging distance L shown in FIG. **7**. The flight altitude H is calculated by correcting a detection value of the atmospheric pressure sensor **32** shown in FIG. **2** using a relative height. The flight altitude may store a value based on GPS positioning acquired by using the GPS receiver **30** and a value based on the detection value of the atmospheric pressure sensor **32**.

(115) For example, it is possible to acquire the flight altitude H based on GPS positioning immediately after the imaging drone **12** takes off, and acquire the corrected flight altitude H using the detection value of the atmospheric pressure sensor **32** during the flight of the imaging drone **12**.

(116) θ is a depression angle of the camera **14**, and is calculated based on the attitude information of the imaging drone **12** and the attitude information of the camera **14**. Reference numeral P.sub.1 represents an imaging position of the imaging drone **12**. A coordinate value of the imaging position P.sub.1 is calculated based on latitude information and longitude information of the imaging drone **12**.

(117) A point P.sub.2 is a point where the optical axis A of the imaging lens **14B** intersects the ground G. A point P.sub.3 is one end of the imaging range. A point P.sub.4 is the other end of the imaging range.

(118) A coordinate value of the point P.sub.3 and a coordinate value of the point P.sub.4 can be calculated by applying the parameters shown in FIG. **8** to the geometric operation. A length β of any one side constituting the imaging range is calculated as a distance between the point P.sub.3 and the point P.sub.4.

(119) The imaging range of the camera **14** provided in the imaging drone **12** during flight on the ground G has a trapezoidal shape in which a base on a side close to the camera **14** is relatively short and a base on a side far from the camera **14** is relatively long. The ground referred to here is a surface that serves as a reference for the altitude of the imaging drone **12**.

(120) The imaging range of the camera **14** can be defined by using the coordinate values of the two-dimensional coordinate system defined in the survey range map. For example, the imaging range of the camera **14** can be defined by using the coordinate values of the respective apexes of the trapezoidal shape representing the imaging range.

(121) The imaging range of the camera **14** can be defined by using the position information of the imaging drone **12**. For example, the latitude and longitude of each apex of the trapezoidal shape representing the imaging range may be calculated based on the latitude and longitude of the

imaging drone **12**.

(122) [Specific Example of Survey Range Map]

(123) FIG. **9** is an explanatory diagram of a survey range map. FIG. **9** shows a survey range map **200** corresponding to a part of the survey range SA in accordance with a size of the display **18A** of the terminal device **18**. An imaging range **204** of the camera **14** and position information **206** of the imaging drone **12** are superimposed on the survey range map **200**. Coordinate values in the survey range map **200** are defined for an apex P.sub.11, an apex P.sub.12, an apex P.sub.13, and an apex P.sub.14 constituting the imaging range **204** of the camera **14**. Similarly, coordinate values in the survey range map **200** are defined for the position information **206** of the imaging drone **12**. Note that the imaging range **204** corresponds to the imaging range F shown in FIG. **1**.

(124) FIG. **9** shows an aspect in which an image simulating the imaging drone **12** is applied as the position information **206** of the imaging drone **12**, but any image, text information, or the like can be applied to the position information **206** of the imaging drone **12**. The position information **206** of the imaging drone **12** described in the embodiment is an example of the camera position information.

(125) In a case in which an operator, such as an operator of the imaging drone **12**, touches any position of the display **18A**, the terminal device **18** may enlarge and display the survey range map **200** with a position of the survey range map **200** corresponding to the touched position as a center.

(126) In addition, in a case in which the operator scrolls any position of the display **18A**, the terminal device **18** may change a region of the survey range map **200** to be displayed on the display **18A** in response to the scrolling of the operator.

(127) That is, the terminal device **18** can change a display aspect of the survey range map **200** to be displayed on the display **18A** in response to the operation of the operator in a case in which the operator operates the terminal device **18**. The change in the display aspect of the survey range map **200** can include enlargement, reduction, shift, and the like.

(128) FIG. **10** is an explanatory diagram of a survey range map on which a not-yet-imaged-house image is superimposed. On a survey range map **220** shown in FIG. **10**, a survey target house image is superimposed as the survey target house information. The survey target house images are classified into a not-yet-imaged-house image **210** representing a not-yet-imaged house and an imaged-house image **212** representing an imaged house.

(129) A polygonal shape corresponding to an outer peripheral shape of the survey target house is applied to the survey target house image. Coordinate values in the survey range map **200** are defined for an apex P.sub.21, an apex P.sub.22, an apex P.sub.23, and an apex P.sub.24 constituting the survey target house image. The coordinate values of the apexes constituting the survey target house image are applied to determination as to whether or not the survey target house is an imaged house or a not-yet-imaged house. In FIG. **10**, the apexes constituting the optional not-yet-imaged-house image **210** which is the survey target house image are denoted by reference numerals P.sub.21, P.sub.22, P.sub.23, P.sub.24, P.sub.25, P.sub.26, and P.sub.27 representing the apexes.

(130) The not-yet-imaged house is a survey target house that is not determined to be included in the imaging range **204** of the camera **14**. The imaged house is a survey target house that has been determined to be included in the imaging range **204** of the camera **14** at least once.

(131) Examples of a building that can be a survey target house include a building listed in a basic resident register among buildings in each municipality. That is, the survey target house is defined by the local government where the housing damage certification survey is conducted.

(132) In the survey range map **200** before the imaging range **204** is decided in the first imaging, the not-yet-imaged-house image **210** may be created for all the survey target houses, and the not-yet-imaged-house images **210** corresponding to all the survey target houses may be superimposed on the survey range map **200**.

(133) The determination of the imaged house and the determination of the not-yet-imaged house are performed in accordance with the decision of the imaging range **204**, and the not-yet-imaged-

house image **210** and the imaged-house image **212** are superimposed on the survey range map **220** according to the determination result.

(134) FIG. **11** is an explanatory diagram of transition of a survey range map. FIG. **11** shows transition of a survey range map in a case in which imaging of the survey range is performed at a first imaging position and a second imaging position that are different from each other. A survey range map **230** is generated based on imaging of a survey range at the first imaging position.

Position information **206A** of the imaging drone **12** is superimposed on the first imaging position.

(135) The imaged house determination unit **88** shown in FIG. **4** determines a not-yet-imaged house included in an imaging range **204A** in the survey range map **230** as an imaged house. The not-yet-imaged house determination unit **90** determines a not-yet-imaged house of which coordinate values of one or more apexes in the not-yet-imaged-house image **210** are not included in the imaging range **204A**, as a not-yet-imaged house that is not included in the imaging range **204A**.

(136) Since all apexes of the not-yet-imaged-house image **210A** shown in FIG. **11** are included in the imaging range **204A**, a not-yet-imaged house corresponding to the not-yet-imaged-house image **210A** is determined as an imaged house. The imaged-house image **212** having the same outer shape as the not-yet-imaged-house image **210A** is superimposed on the house determined as an imaged house, instead of the not-yet-imaged-house image **210A**.

(137) A survey range map **240** is generated based on imaging of a survey range at the second imaging position. For example, the second imaging position is an imaging position where imaging of the survey range is performed next to the first imaging position. Position information **206B** of the imaging drone **12** is superimposed on the second imaging position.

(138) In the survey range map **240**, the imaged-house image **212** is superimposed on the position of the not-yet-imaged-house image **210A** determined to be an imaged house in the survey range map **230**. A display aspect that is visually distinguishable from the not-yet-imaged-house image **210** is applied to the imaged-house image **212**. As an example of the display aspect applied to the imaged-house image **212**, an aspect in which a color, a brightness, a pattern, and the like are differently set with respect to the not-yet-imaged-house image **210** can be applied.

(139) The imaged house determination unit **88** shown in FIG. **4** determines a not-yet-imaged house included in an imaging range **204B** in the survey range map **240** as an imaged house. The imaged-house image is superimposed on the position of the not-yet-imaged house determined as an imaged house, instead of the not-yet-imaged-house image **210B**.

(140) In this way, the imaging drone **12** is moved, the imaging position in the survey range is sequentially changed, and the imaged-house image **212** is superimposed on the survey target house that is included in the imaging range **204** of the camera **14** at least once, as the imaged house.

(141) [Modification Example of Survey Range Map]

(142) FIG. **12** is an explanatory diagram of a survey range map according to a modification example. A survey range map **250** shown in FIG. **12** includes navigation information. In FIG. **12**, a first navigation image **252**, a second navigation image **254**, and a third navigation image **256** are illustrated as the navigation information.

(143) The first navigation image **252** is an arrow line representing a direction in which a not-yet-imaged house is located with respect to the position of the imaging drone **12**. In addition, the first navigation image **252** includes text information representing the number of the not-yet-imaged houses. As the number of the not-yet-imaged houses applied to the first navigation image **252**, the number of the not-yet-imaged houses included in a range having a predetermined distance from the position of the imaging drone **12** is applied.

(144) In the survey range map **250** shown in FIG. **12**, the first navigation image **252** is disposed for each of the north, south, east, and west. The survey range map **250** may include the first navigation image **252** facing northeast, the first navigation image **252** facing southwest, and the like. In the survey range map **250**, the first navigation image **252** in which the number of the not-yet-imaged houses is zero may be deleted.

(145) The second navigation image **254** includes an arrow line representing a direction of the not-yet-imaged house of interest at the position of the imaging drone **12** and text information representing a distance from the position of the imaging drone **12** to the not-yet-imaged house of interest.

(146) As the distance from the position of the imaging drone **12** to the not-yet-imaged house of interest, a distance from the position of the imaging drone **12** to a position on a flight path of the imaging drone **12** corresponding to the position of the not-yet-imaged house of interest can be applied.

(147) A wide area map showing the entire survey range SA is applied to the third navigation image **256**. In the third navigation image **256**, a frame **258** representing a region corresponding to the survey range map **250** is superimposed on the wide area map.

(148) The first navigation image **252** described in the embodiment is an example of the first navigation information. The second navigation image **254** described in the embodiment is an example of the second navigation information. The third navigation image **256** described in the embodiment is an example of the third navigation information.

(149) [Operations and Effects of Information Processing System and Information Processing Method According to Embodiment]

(150) The information processing system and the information processing method according to the embodiment can obtain the following operations and effects. [1]

(151) The imaging range **204** of the camera **14** provided in the imaging drone **12** on the ground G and the not-yet-imaged-house image **210** representing the not-yet-imaged house are superimposed on the survey range map **220** representing the geographical information of the survey range. The survey range map **220** on which the not-yet-imaged-house image **210** is superimposed is displayed on the display **18A** of the terminal device **18**.

(152) Accordingly, the operator of the imaging drone **12** can recognize the imaging range of the camera **14** during flight and the not-yet-imaged house, and can cause the imaging drone **12** to fly toward the not-yet-imaged house. [2]

(153) The imaged-house image **212** representing the imaged house is superimposed on the survey range map **220**. The survey range map **220** on which the imaged-house image **212** is superimposed is displayed on the display **18A** of the terminal device **18**. Accordingly, the operator of the imaging drone **12** can distinguish between the not-yet-imaged house and the imaged house. [3]

(154) The position information **206** of the imaging drone **12** is superimposed on the survey range map **220**. Accordingly, the operator of the imaging drone **12** can grasp the positions of the imaging drone **12** and the camera **14**. [4]

(155) The first navigation image **252** representing the direction in which the not-yet-imaged house is located is superimposed on the survey range map **250**. Accordingly, the operator of the imaging drone **12** can grasp the direction in which the not-yet-imaged house is located. [5]

(156) The first navigation image **252** includes the number of the not-yet-imaged houses in the direction represented by the first navigation image **252**. Accordingly, it is possible to guide the imaging drone **12** according to the number of the not-yet-imaged houses. [6]

(157) The second navigation image **254** representing the direction from the camera **14** to the not-yet-imaged house of interest and the distance from the camera **14** to the not-yet-imaged house of interest is superimposed on the survey range map **250**. Accordingly, the operator of the imaging drone **12** can grasp the direction and the distance from the camera **14** to the not-yet-imaged house of interest. [7]

(158) The third navigation image **256** including the wide area map showing the entire survey range is superimposed on the survey range map **250**. Accordingly, the operator of the imaging drone **12** can grasp the position of the imaging drone **12** in the entire survey range. [8]

(159) A not-yet-imaged house of which all apexes constituting the not-yet-imaged-house image **210A** are included in the imaging range **204A** is determined as an imaged house. Accordingly, it is

possible to suppress the occurrence of omission in the captured image of the imaged house.

(160) A history of determination results is stored in a case in which all the survey target houses are determined as imaged houses. Accordingly, it is possible to avoid duplication of imaging for the same survey target house.

(161) [Modification Example of Information Processing System According to Embodiment]

(162) A part of the information processing performed in the terminal device **18** may be performed by other devices of the terminal device **18**, such as the controller **16** and the server **20**. For example, the controller **16** may perform the estimation of the imaging range of the camera **14** on the ground G.

(163) The terminal device **18** can acquire the survey range map **220**, the position of the survey target house, and the size of the survey target house from the server **20**, generate the not-yet-imaged-house image **210** in accordance with the performance of the imaging by the camera **14**, and superimpose the not-yet-imaged-house image **210** on the survey range map **220**. In addition, the terminal device **18** may acquire the survey range map **220** in which the two-dimensional coordinate system is set in advance.

(164) In the embodiment of the present invention described above, the configuration requirements can be changed, added, or deleted as appropriate without departing from the spirit of the present invention. The present invention is not limited to the embodiment described above, and many modifications can be made by a person having ordinary knowledge in the art within the technical idea of the present invention. In addition, the embodiment, modification example, and application example may be combined as appropriate.

EXPLANATION OF REFERENCES

(165) **10**: information processing system **12**: imaging drone **12A**: main body **12B**: rotor **13**: gimbal **14**: camera **14A**: image sensor **14B**: imaging lens **16**: controller **18**: terminal device **18A**: display **20** server **22**: network **30** GPS receiver **32**: atmospheric pressure sensor **34**: azimuth sensor **36**: gyro sensor **38**: communication interface **50** user interface **52**: imaging setting unit **54**: flight control unit **56**: sensor data storage unit **58**: image data storage unit **60** imaging control unit **62**: communication interface **80** survey range map acquisition unit **82**: camera attitude information acquisition unit **83**: camera configuration information acquisition unit **84**: imaging range calculation unit **86**: survey target house information acquisition unit **88**: imaged house determination unit **90** not-yet-imaged house determination unit **92**: survey range map processing unit **94**: not-yet-imaged-house image generation unit **96**: imaging-range image generation unit **98**: imaged-house image generation unit **99**: display control unit **100**: processor **102**: computer-readable medium **104**: communication interface **106**: input/output interface **108**: bus **110**: input device **120**: image processing program **122**: storage control program **124**: communication control program **126**: display control program **200**: survey range map **204**: imaging range **204A**: imaging range **204B**: imaging range **206**: position information of imaging drone **12** **206A**: position information of imaging drone **12** **210**: not-yet-imaged-house image **210A**: not-yet-imaged-house image **210B**: not-yet-imaged-house image **212**: imaged-house image **220**: survey range map **230**: survey range map **240**: survey range map **250**: survey range map **252**: first navigation image **254**: second navigation image **256**: third navigation image **S10** to **S28**: each step of information processing method

Claims

1. An information processing apparatus comprising: a processor; and a memory in which a program to be executed by the processor is stored, wherein the processor executes a command of the program to acquire position and attitude information representing a position of a camera mounted on a flying object and an attitude of the camera, acquire camera configuration information including an optical configuration of the camera, calculate an imaging range of the camera based

on the position and attitude information and the camera configuration information, acquire a survey range map showing a survey range, acquire survey target house information including a position of a survey target house in the survey range map, determine whether or not the survey target house is included in the imaging range, specify a not-yet-imaged house that does not have a history of being determined to be included in the imaging range, and superimpose information on the not-yet-imaged house on the survey range map.

2. The information processing apparatus according to claim 1, wherein the processor superimposes, on the survey range map, a not-yet-imaged-house image corresponding to the not-yet-imaged house.

3. The information processing apparatus according to claim 1, wherein the processor superimposes, on the survey range map, information on an imaged house having a history in which the survey target house is determined to be included in the imaging range.

4. The information processing apparatus according to claim 3, wherein the processor superimposes, on the survey range map, an imaged-house image corresponding to the imaged house.

5. The information processing apparatus according to claim 1, wherein the processor superimposes, on the survey range map, camera position information representing a position of the camera in the survey range map.

6. The information processing apparatus according to claim 1, wherein the processor superimposes, on the survey range map, first navigation information representing a direction in which the not-yet-imaged house is located.

7. The information processing apparatus according to claim 6, wherein the processor superimposes, on the survey range map, the first navigation information including information representing the number of the not-yet-imaged houses in the direction represented by the first navigation information.

8. The information processing apparatus according to claim 1, wherein the processor superimposes, on the survey range map, second navigation information representing a distance and a direction from the imaging range to the not-yet-imaged house of interest.

9. The information processing apparatus according to claim 1, wherein the processor superimposes, on the survey range map, third navigation information including a wide area map in which an entirety of the survey range map is reduced.

10. The information processing apparatus according to claim 1, wherein the processor acquires the position of the camera, an altitude of the camera, an orientation of an optical axis of the camera, and a depression angle of the optical axis, as the position and attitude information.

11. The information processing apparatus according to claim 1, wherein the processor determines whether or not the not-yet-imaged house is included in the imaging range in a case in which the imaging range moves according to movement of the flying object.

12. The information processing apparatus according to claim 1, wherein the processor stores a history in which the survey target house is determined to be included in the imaging range.

13. An information processing method executed by a computer, the method comprising: acquiring position and attitude information representing a position of a camera mounted on a flying object and an attitude of the camera; acquiring camera configuration information including an optical configuration of the camera; calculating an imaging range of the camera based on the position and attitude information and the camera configuration information; acquiring a survey range map showing a survey range; acquiring survey target house information including a position of a survey target house in the survey range map; determining whether or not the survey target house is included in the imaging range; specifying a not-yet-imaged house that does not have a history of being determined to be included in the imaging range; and superimposing information on the not-yet-imaged house on the survey range map.

14. A non-transitory, computer-readable tangible recording medium on which a program for causing, when read by a computer, the computer to execute the information processing method

according to claim 13 is recorded.

15. An information processing system comprising: a flying object; a camera mounted on the flying object; and an information processing apparatus equipped with a processor and a memory in which a program to be executed by the processor is stored, wherein the processor executes a command of the program to acquire position and attitude information representing a position of the camera mounted on the flying object and an attitude of the camera, acquire camera configuration information including an optical configuration of the camera, calculate an imaging range of the camera based on the position and attitude information and the camera configuration information, acquire a survey range map showing a survey range, acquire survey target house information including a position of a survey target house in the survey range map, determine whether or not the survey target house is included in the imaging range, specify a not-yet-imaged house that does not have a history of being determined to be included in the imaging range, and superimpose information on the not-yet-imaged house on the survey range map.
